Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Illustrate the Lloyd-Max algorithm for scalar quantization, assuming full knowledge of the source pdf. How does the rate-distortion performance of Lloyd-Max compare with other conventional quantizers (uniform, ECSQ, VQ)?
Explain how the Lloyd-Max quantizer can be designed when the pdf of the source is unknown but a training set of N samples is given.

- Write the pseudocode of the algorithm in this case
- Simulate the execution of the algorithm generating $N = 4$ samples at random in the interval $[0, 1]$.

Solution

- For a description of L-M, see slide set `MMSP_1.2_Lossy coding`. Let $D_{LM}(R)$ denote the rate-distortion curve for the Lloyd-Max quantizer and a given pdf. When distortion is expressed in terms of MSE, the following inequality holds

$$D_{VQ} < D_{ECSQ} \leq D_{LM} \tag{1}$$

  where the last inequality holds with the equality sign when the output symbols of the Lloyd-Max quantizer are equally likely.

- Let $\{x_1, x_2, \ldots, x_N\}$ denote a set of samples drawn from an unknown pdf. Let M denote the number of intervals, a_k, $k = 1, \ldots, M+1$, the thresholds and y_k, $k = 1, \ldots, M$, the centroids.

  1. Randomly initialize the thresholds a_k, $k = 2, \ldots, M$
  2. For each interval $I_k = [a_k, a_{k+1})$, $k = 1, \ldots, M$, compute the centroid

$$y_k = \frac{1}{N_k} \sum_{x_i \in I_k} x_i \tag{2}$$

     where N_k is the number of samples in I_k
  3. Compute the thresholds as

$$a_k = \frac{y_{k-1} + y_k}{2} \tag{3}$$

  For example, let $\{0.1, 0.2, 0.6, 0.8\}$ denote a set of four samples and set $M = 2$.

  - Randomly initialize the thresholds. $a_1 = -\infty$, $a_2 = 0.65$, $a_3 = +\infty$

- $y_1 = \frac{0.1+0.2+0.6}{3} = 0.3$, $y_2 = 0.8$
- $a_2 = \frac{0.3+0.8}{2} = 0.55$
- $y_1 = \frac{0.1+0.2}{2} = 0.15$, $y_2 = \frac{0.6+0.8}{2} = 0.7$
- $a_2 = \frac{0.15+0.7}{2} = 0.425$
- $y_1 = \frac{0.1+0.2}{2} = 0.15$, $y_2 = \frac{0.6+0.8}{2} = 0.7$
- Convergence detected. Stop

2 Question - 5 pts

Illustrate the basic principles behind the MDCT. Illustrate how the MDCT is able to accommodate signals with non-stationary time-frequency characteristics.

Solution

See slide set `MMSP_2.2_Audio coding coding`. To answer the second part of the question, you are asked to discuss the use of variable size windows.

3 Question - 12 pts [Matlab]

Consider a memoryless source, uniformly distributed in the interval $[-1, 1]$.

- Design a suitable quantizer for the source at hand.
- For each target bitrate R [bps], *empirically* measure the output distortion of the uniform scalar quantizer.
- Plot the rate-distortion functions of the quantizer.
- Compare the plots with the theoretical Shannon's bound.

Solution

```
clear all
close all
clc

%Create source
N = 10^5;
x = 2*rand(N,1)-1;
Xmin = -1;
Xmax = 1;

%the optimal quantizer for a uniform distributed source is a uniform
%quantizer!
R = 1:8;
MSE = NaN*ones(size(R));

ir = 1;
for r = R
    STEP = (Xmax - (Xmin))/2^r;
    x_q =  STEP*floor(x./STEP) + STEP/2;
    MSE(ir) = var(x - x_q);
    ir = ir+1;
end

SNR = 10*log10(var(x)./MSE);
plot(R,SNR,'-ob');

%Shannon bound:

%D(R) = sigma^2_x * 2^(-2R)
%the SNR is then 10*log10( sigma^2_x / sigma^2_x * 2^(-2R) )
```

```
%which is 2*R * 10 * log10(2) = 6.02*R

D = 2*R* 10*log10(2);
hold on
plot(R,D,'r');
xlabel('rate [bps]');
ylabel('SNR [db]');
title('Rate-distortion curve');
legend('Empirical', 'Shannon bound','location','southeast');
```

4 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a discrete memoryless source with finite alphabet, and $D(R)$ the corresponding rate-distortion curve
   - [T] [F] $D(0) = H(X)$
     FALSE. $D(0) = \sigma_x^2$.
   - [T] [F] If $\langle D, R\rangle$ is achievable, than $\langle D, R + \Delta R\rangle$, $\Delta R > 0$ is achievable
     TRUE. For a target distortion, I can always spend more bits than for an achievable point.
   - [T] [F] $R(0) = \infty$
     FALSE. Lossless coding a discrete memoryless source requires that $R(0) \geq H(X)$.
   - [T] [F] $H(X) \leq \log_2(M)$, where M is the alphabet size
     TRUE. With equality when symbols are equally likely.
2. Consider a speech coding system
   - [T] [F] The long term predictor is useful when coding unvoiced speech
     FALSE. For unvoiced speech, only short-term correlations can be exploited.
   - [T] [F] Direct quantization of prediction coefficients might lead to instability at the decoder
     TRUE. Direct quantization might cause the poles of the shaping filter to be outside the unit circle, thus causing instability at the decoder side.
   - [T] [F] The number of poles of the whitening filter $A(z)$ depends on the number of formants below 4 kHz
     FALSE. The number of **zeros** depends on the number of formants below 4 kHz, as $A(z)$ is an all-zero filter.
3. Consider a video coding system. Given a video sequence of fixed spatio-temporal resolution:
   - [T] [F] The computational complexity of motion estimation does not depend on the block size.
     TRUE. In a $N \times N$ frames there are N^2/B^2 blocks of size $B \times B$. The number of operations is $O(C \cdot \frac{N^2}{B^2} B^2) = O(C \cdot N^2)$, where C is the number of candidates tested per block.
   - [T] [F] The coding efficiency increases when increasing the GOP size (e.g. the time interval between I-frames).
     TRUE. For a target level of distortion, $R_I > R_P$. The longer the GOP size, the larger is the fraction of I-frames.
   - [T] [F] Entropy coding is the most computationally intensive operation.
     FALSE. Motion estimation is the most computationally intensive operation.